

T1 SIGNALLING 信号编码		
encoding	clock recovery	BW
NRZ	none	high
NRZI (flip on 1)	1s only	high
Manchester (mid-bit flip)	self-clocking [自同步]	×2

Modulation: AM=amplitude / FM=frequency / PM=phase. **QAM**=amp+phase, bits/symbol=log₂(M). 35-QAM legal but wasteful (5.1bit).
Capacity: Nyquist (noiseless) / Shannon (with noise). Narrow pulse ↔ wide spectrum [时窄频宽]. **dB**=ratio ×10log; **dBm** zero = 1mW; **dBi** = gain vs isotropic antenna.
Circuit vs packet: circuit dedicated, QoS guaranteed, but wastes idle + single-point failure [独占浪费]; packet = stat-mux, efficient but jitter/reorder.

T2 MEDIA 铜·光·无线		
Copper foes: attenuation / skin effect (HF ↑R) / antenna effect (crosstalk). Fix = differential signalling + twisting. DSL = DMT multi-carrier (4.3kHz sub-channels × adaptive QAM). [频高距短] higher freq → shorter reach. ADSL ~1MHz/4km → G.fast ~100MHz/100m.		
fibre	core	reach
multimode step/graded	50–62µm	~1–2km [模态失真]
single-mode SMF	~9µm	50–100km

Total internal reflection; ~0.2dB/km; **WDM/DWDM** (80–160λ ×100G = 16Tbps); EDFA every 80–100km amplifies signal+noise; OEO regen = clean but costly.
Wireless: low freq = far/thru-wall/slow; high freq = near/fast. [远·快·低功耗三选二] P ∝ 1/d². Multipath → fading. Regulator = ACMA.
FTTx [铜段越短速越快]: FTTP (no copper) > FTTC 10-100m > FTTN 100-600m > FTTB (in-building); HFC = shared coax (DOCSIS); PON = passive splitter 1:32/64.

T3 LAN / ETHERNET / WIFI		
Key concepts: Administrative domain (one ruleset/owner) [同规则]. Broadcast domain (broadcast reach) [广播可达]. LAN philosophy = simple + fast, complexity pushed up (best-effort, CRC only). Eth frame: Preamble7 SFD1 Dst6 Src6 VLAN4 Type2 Payload46-1500 FCS4 Type 0x0800=IPv4. MAC=48bit, broadcast FF×6, first 24=OUI. Jumbo=9000B.		
MAC proto	scheme	util
Pure ALOHA	send anytime	~18%
Slotted ALOHA	slot-aligned	~37%
CSMA/CD	wired, listen-while-send	—
CSMA/CA	wireless, avoid+ACK	—

min frame time ≥ 2× one-way prop delay [碰撞窗口] (Eth=64B), else collision undetectable. CD backoff = binary exponential (0~2ⁿ-1), give up at 16.
Wireless [有线CD vs 无线CA]: hidden terminal (A/C unheard → collide at B) → **RTS/CTS**; exposed terminal (false busy). WiFi: per-frame ACK, no CD.
2.4GHz non-overlap only 1/6/11 [仅3个], crowded but thru-wall; **5GHz** more channels, less interference; 6GHz cleanest.
Switch: learn src MAC, unknown → flood, broadcast → always flood. **STP** (802.1D) prunes to loop-free tree [防广播风暴] (root → root-port → block redundant). **VLAN** 802.1Q tag 4B, VID 0-4095.

T4 IP / TRANSPORT		
PDU + device per layer [别混用] Packet≠Frame≠Segment:		
layer	PDU	device
App	Message/Data	Proxy, Gateway
Transport	Segment/Datagram	—
Network	Packet	Router
Link	Frame	Switch, Bridge
Physical	Bits	Hub, Repeater

SDU = payload a layer receives from the one above [上层数据=本层载荷]. e.g. TCP's SDU = HTTP data; IP's SDU = TCP segment.
Layering: separation / abstraction / interop. Encapsulation = each layer adds header, reads only its own. **End-to-end:** smart edge, dumb core.
[Frame逐跳重封装·Packet端到端不变] (except TTL).
IPv4 hdr 20B (IHL≥5×4): Ver·IHL·DSCP·ECN·TotLen | ID·DF·MF·FragOff | TTL·Proto·HdrChksum | Src | Dst
Proto: 1=ICMP, 6=TCP, 17=UDP. TTL -1/hop → 0 drop; changing TTL means recompute checksum [IPv6无此].

special addr	meaning
127/8	loopback
169.254/16	link-local [DHCP失败]
10/8·172.16/12·192.168/16	private RFC1918
224/4	multicast (IGMP joins)

Longest prefix match: pick most specific (/24 over /16) [最具体优先]. **Forwarding** (local lookup·data plane) vs **Routing** (global table build·control plane) [本地vs全局].
ARP: IP → MAC within LAN (bcast req·ucast reply·cache); Gratuitous ARP detects conflict/announces.
ICMP (Proto1): 0 EchoReply·8 EchoReq(ping)·11 TimeExceeded (traceroute = TTL increment) [TTL递增]·3 DestUnreach.
Fragment: split if >MTU, same ID, MF(more), Offset(×8B); **receiver reassembles** [接收端重组·一片丢全废]. **PMTUD:** DF=1 + ICMP probes min MTU. MTU: Eth1500/WiFi~2300/PPPoE1492.
IPv6: 128bit, fixed 40B hdr, **no checksum / no broadcast / source-only fragment** [三大变化], + Anycast (nearest member, DNS root), SLAAC. NAT = "IPv6 killer".

TCP vs UDP		
	UDP	TCP
conn	none	3-way handshake
reliable/order	no	ACK+retx / ordered
flow/cong	no	rwnd / AIMD
header	8B	20+B
use	DNS·RTP·games	HTTP·files

Handshake SYN → SYN-ACK → ACK [各确认收发]. Cumulative ACK. Socket = 5-tuple (proto,srcIP,srcPort,dstIP,dstPort). **NAPT** uses port to map inside hosts. Port = 16bit.

T5 DNS / DHCP	
DNS = distributed / hierarchical / automated. root . → TLD → 2nd-level. Domain (admin boundary) vs Zone (data file) vs Delegation [边界vs数据].	
RR	role
SOA	zone start (one/zone)
A/AAAA	→ IPv4/IPv6
CNAME	alias
MX	mail (w/ priority)
NS	authoritative nameserver

Recursive (resolver does all) vs **iterative** (NS only points next) [客户端vsresolver]. **UDP/53** default, large → TCP/53. 13 root IPs (A-M) + Anycast. TTL caches.
DHCP DORA: Discover(bcast) → Offer → Request(bcast, so losers reclaim) [让落选回收] → Ack(lease).

T5 HTTP / WEB	
Methods GET/HEAD/POST. Status 2xx OK·3xx redirect·4xx client(404)·5xx server. Stateless → Cookie/Token [无状态]. 3rd-party cookie = cross-site tracking. Perf: parallel (many TCP, costly) · persistent (1.1 reuse) · pipelining (HoL blocking) [队头阻塞]. Cache: ETag/If-None-Match, Last-Modified/If-Modified-Since (Conditional GET) [没变不传]. Proxy = cache-on-pull; CDN = push + DNS to nearest [拉vs推]. HTTP/3 = QUIC/UDP, kills TCP HoL.	

T5 REALTIME	
best-effort net, 3 problems: Latency (overall slow) / Jitter (variance ~15ms) / Loss . [到太晚=丢] equivalent loss = arrives past playout. Buffer: big=low-loss/high-delay, small=opposite. Fixes: retransmit (slow, rare here) · elastic buffer · FEC (send redundancy R=P1⊕P2⊕P3 ahead) [带宽换延迟] · codec adapt. RTP (media, even port) / RTCP (feedback, odd port) / SIP (signalling) / DASH (HTTP adaptive).	

T5 MQTT / IoT		
Pub/Sub decoupled: publisher → broker(1883) → subscriber, never direct [解耦]. Compact binary, fixed hdr ≥2B. + = one level, # = this level & below (end only, skips \$SYS).		
QoS	pkts	guarantee
0 fire&forget	1	at most once
1 PubAck	2	at least once (dup)
2 Rec/Rel/Comp	4	exactly once

effective QoS = min(pub, sub) [取小], hop-by-hop not end-to-end. **Retain** = last value for new subs (empty msg clears) [给新订阅者]. **LWT** = will on disconnect. Clean vs Persistent session (replays missed).

T5 P2P / DHT	
BitTorrent: piece 256/512KB, sub-piece 16KB; Rarest First [优先最稀有] + Random start + Endgame; Choking: re-eval 10s, optimistic unchoke 30s, ~4 concurrent; Tracker gives peers not files; SHA-1. Seeder ranks by upload rate. DHT (replaces Tracker): ring, KeyID → nearest successor NodeID (160bit), finger table → O(log N) hops ; replicate to pred/succ vs churn. exact lookup only, no wildcard/regex [结构锁死查询] (hash kills order) — same issue as MQTT wildcards.	

GUEST / GOPHER	
Gopher (1991, TCP/70): minimal, selector → menu text + lone "." end; type 1 dir/0 text/7 search. Server-defined tree vs HTML hyperlink graph [准定义结构]. Free Web (1993) beat paid Gopher. Guest (Geoff Huston): IPv6 deploy ~40%+; bgp.potaroo.net for global BGP state.	

EXAM TRAPS 最易错

- **BGP prepending** → longer path → less preferred [方向别反] (for backup link). ≠ path injection (attack).
- **effective QoS = min(pub, sub)** [取小]; MQTT is hop-by-hop, not e2e.
- **Fast Retransmit ≠ Fast Recovery**: FR = when to resend (3 dup-ACK); FR recovery = no Slow Start, MD only. **Tahoe has FR, no Fast Recovery; Reno has both** [易混].
- **WIN (rcv buffer) ⊥ CWND (net estimate)** [完全无关]. effective = min(W,WIN,CWND).
- **Counter equal = trustworthy; Gauge equal = NOT** [计量陷阱]. Counter32 wraps at 4.29B; 10Gbps wraps ~3.4s → Counter64.
- **SNMP: agent = server, manager = client** [反直觉].
- **Hierarchical routing "hop" unit changes** [量纲变了]: routers → regions, 6>5 not worse.
- **Onion routing = telescoping** [逐跳建链], not 3 keys at once; timing analysis defeats encryption.

T6 ROUTING

Optimality property: sub-paths of a shortest path are also shortest [子路径也最短]. **Sink/Source tree**: identical if cost symmetric, diverge if not [对称时一致]. Unicast / Multicast (224/8 group) / Broadcast (domain only).

	Link-State	Dist-Vector
algo	Dijkstra	Bellman-Ford
knows	full topo (flood LSP)	neighbours + dist
converge	fast	slow (rounds)
scale	OK	better
proto	OSPF/IS-IS (intra)	RIP (∞=16)

Mnemonic: Dijkstra trades global knowledge for convergence speed; Bellman-Ford trades convergence quality for scalability [口诀].

Count-to-infinity (bad news travels slow) [坏消息慢] → Split Horizon / Poison Reverse. **ECMP** = equal-cost multipath (hash per flow). **Hierarchical**: aggregation shrinks tables, cost = suboptimal path [缩表换次优].

T6 BGP 域间·策略

Runs over **TCP/179**, manually configured peers, **Path Vector** (full AS path, not distance). AS Path: (1) loop detect — own ASN → drop; (2) policy scoring.

Prepending: repeat own ASN to look longer [更长=更不优] → steer to backup. **eBGP** (inter-AS): prepend ASN + rewrite next-hop at boundary. **iBGP** (intra-AS): sync external routes, full-mesh, iBGP-learned not re-advertised over iBGP; OSPF/IS-IS resolves next-hop.

Transit (paid, full reach) vs **Peering** (free, no transit) [付费vs互通]. **Hot potato** = dump early → asymmetric paths. **Local-Pref** = main outbound policy lever [出向主战场]. Control plane right → left vs data left → right [方向相反]. RIB → FIB. No native security (RPKI ~50%, BGPsec ≈ 0).

T7a CONGESTION / TCP

BDP = 2×BW×delay = in-flight data [在途数据]. 3 windows: **W** (sender, BDP) · **WIN** (rcv buffer) · **CWND** (net estimate), send = min(all). Congestion collapse: retx floods → goodput → 0. **AIMD** = additive-up (+1/RTT) · multiplicative-down (×0.5) [慢加快减] → sawtooth. **Slow Start**: CWND=1, double/RTT (actually exponential) [名字反了], loss → ssthresh=½ → AI.

recovery	mechanism
Fast Retransmit	3 dup-ACK → resend that seg
Fast Recovery	no Slow Start, MD only
NewReno	partial ACK → multi per RTT
SACK	exact received, ≤4 blocks

Cong signals: loss (too late) / delay (jitter-sensitive) / **ECN** (router marks IP hdr, zero extra traffic, needs full-path support) [需全路径支持]. ACK clocking = self-clock. Variants: Tahoe/Reno/Vegas (starved by Reno) [被饿死]/CUBIC (Linux def)/BBR (Google, BW+RTT).

T7b MEASUREMENT / SNMP

End-to-end test (speedtest) can't tell which hop fails → query devices directly. Monitoring must not burden device (Seattle router: 60s poll dropped 90% traffic) [别加负担].

part	role
Agent	Server (monitored device)
Manager	Client (NMS)
MIB	data model (ASN.1/OID)

Over **UDP**; agent on **161**, manager **162** (traps). Agent very light: no rate / no history / no absolute clock (TimeTicks=1/100s) [极轻].

5 PDU: Get/GetNext/Set/Response/Trap. **6 Trap**: linkDown/Up · **coldStart** (unexpected reboot) / warmStart (planned) [意外vs预期] · authFail · **egpNeighbourLoss** (BGP peer gone, may go offline) [可能失联].

Counter (up only, wraps to 0, C32/C64) vs **Gauge** (up/down, latches at max) [满则停]. Rate: Δ<0 = wrap or reboot (check sysUpTime). **GetNext** walks table in lexical order; **GetBulk** (v2) fetches a block (≤64KB UDP).
OID: 1.3.6.1.2.1=mib-2, .4=vendor private. **v1** plaintext community / **v2c** +GetBulk+C64 / **v3** 3 levels: noAuthNoPriv·authNoPriv·authPriv.

T7c CRYPTO

4 properties: Confidentiality · Authentication (+ non-repudiation) · Integrity · Freshness (anti-replay).

	symmetric	asymmetric
keys	1 shared	pub+priv pair
speed	fast (GB/s)	slow 100-10 ⁴ ×
weakness	key distribution	pubkey ownership (PKI)
algo	AES/ChaCha 20	RSA2048/ECC256/DH

Hybrid (=TLS): asymmetric opens, symmetric (AES-GCM) carries data [非对称开局]. **Forward secrecy**: ephemeral DH (g^ab, a/b discarded) → future key leak can't decrypt past [用完即扔]. **Modes**: ECB (never)/CBC (POODLE)/GCM (AEAD, modern) [首选].

PKI: CA signs X.509, ~88 root certs. **Revocation broken** (CRL won't scale) [撤销失效]. **TLS** sits between transport & app, handshake before first HTTP req [握手在前]; UDP = DTLS. Post-quantum: Shor breaks RSA/ECC, AES-256 OK if doubled.

T7c FIREWALL / VPN

firewall	inspects
Stateless	per-packet · IP/port (deny all)
Stateful	tracks flow (NAT reply)
App/DPI	reassemble content (costly, can't see encrypted)

DMZ = 2 firewalls (ext → fw1 → DMZ svc → fw2 → internal). ~80% attacks are internal [内部威胁]. **VPN** = IP-in-IP + IPsec (ESP encrypt + auth). Modes: **Tunnel** (router ↔ router, common) / Transport (host ↔ host) / Mixed (worker → gateway). Hides real src/dst. **TunnelVision**: DHCP option 121 pushes two /1 routes (more specific than /0) → bypasses VPN [架空VPN].

T7c ATTACKS

attack	key / defence
DHCP poison	drain pool + impersonate → set Gateway/DNS. Def: DHCP Snooping
ARP poison	Gratuitous ARP → MitM. Def: DAIR (hard)
BGP hijack	more-specific prefix / path injection → global leak
DNS cache poison	spooft src IP + guess 16-bit ID + race [攻边缘resolver]
SYN flood	no final ACK, drains state. Def: SYN cookies

DNSSEC (integrity + authenticity, NOT confidentiality) [不做机密]; **RRSIG** (signs RR set) · **DNSKEY** (KSK signs ZSK, ZSK signs records) · **DS** (parent → child KSK) · **NSSEC** (leaks zone → NSSEC3) [泄露zone]. Trust anchor = root key (hardcoded).

Amplification [两个乘子]: host mult (botnet/IoT) + packet mult (SMURF / DNS / Memcache). **Spoofing fix** = **Ingress filtering** (ISP only; undeployed — protects others not self) [只护别人]. **DDoS**: CDN + upstream black-hole.

T7c ONION / WIFI-SEC

Onion (Tor): Guard (src not dst)/Middle/Exit (dst not src), **telescoping per-hop key** [逐跳建链], layered. Exit sees dst → still need app crypto. **Timing analysis defeats layered encryption** [时序防不住]. vs MPLS (pins whole path) = more fragile.

WiFi	break
WEP (RC4)	broken in minutes · avoid
WPA (TKIP → CCMP)	WPS flaw
WPA2	KRACK / KR00K
WPA3 (SAE)	Dragonblood (huge compute)

Handshake sends no plaintext password: both derive PTK from MAC+nonce+pwd, exchange result+MIC [不传明文]. Enterprise = 802.1X/EAP/RADIUS (no shared PSK). **AP identity unverifiable** → fake SSID [无法验AP]; downgrade / SSL strip (HSTS defends). Golden rule: **no predictable plaintext** [可预测=被破]. Harvest-then-decrypt: store now, crack later.